# run_sjsdm_tuning_candidates.R

August 11, 2026

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`run_sjsdm_tuning_candidates` *Run sjSDM Tuning Candidates*

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### Description

Fits every deterministic regularization candidate independently on each cross-validation training partition and scores held-out probabilities.

### Usage

```
run_sjsdm_tuning_candidates(
  data_assignments = NULL,
  data_candidates = NULL,
  prepare_fold_function = NULL,
  fit_function = NULL,
  predict_function = NULL,
  score_function = score_sjsdm_tuning_predictions,
  seed = 900723L,
  epsilon = 1e-06,
  retain_prediction_cache = FALSE
)
```

### Arguments

`data_assignments`
: Location-level assignment table with ‘repeat_id‘, ‘fold_id‘, ‘location_id‘, ‘n_samples‘, and ‘row_indices‘. Optional ‘cv_strategy‘ is propagated.

`data_candidates`
: Candidate table returned by [build_sjsdm_regularization_candidates()].

`prepare_fold_function`
: Injectable function called with ‘train_indices‘, ‘test_indices‘, ‘repeat_id‘, and ‘fold_id‘. It must return ‘data_train_input‘, ‘data_test_input‘, and matrix ‘data_test_observed‘.

fit_function Injectable function called with ‘data_train_input‘, one-row ‘candidate‘, and deterministic integer ‘seed‘.

predict_function Injectable function called with fitted ‘object‘ and ‘data_test_input‘. It must return a probability matrix aligned to ‘data_test_observed‘.

score_function Injectable function called with the fitted object, test input, aligned observations and probabilities, epsilon, and, when supported, a deterministic ‘score_seed‘. Defaults to marginal binary prediction scoring; production sjSDM tuning supplies the joint-likelihood scorer.

seed Single non-negative integer used to derive stable fit and score seeds. Defaults to ‘900723L‘.

epsilon Probability clipping tolerance used for held-out log loss. Defaults to ‘1e-6‘.

retain_prediction_cache Logical. When true, retain compact fold metadata and candidate probability matrices so selected OOF artifacts can be assembled without refitting.

## Details

Fold preparation is run once per repeat and fold. Preparation, fit, prediction, and scoring errors are retained as structured status rows so a failed candidate does not abort the remaining tuning grid.

## Value

Compact tibble with one row per repeat, fold, and candidate. It contains candidate parameters, fold counts, total and per-response held-out negative log likelihood, macro AUC, fit status, error text, CV strategy, and regularization source. Fitted objects and prediction matrices are omitted by default. With ‘retain_prediction_cache = TRUE‘, a named list contains the unchanged tuning table and one compact cache entry per fold.

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